

Accelerated gradient descent: momentum

Illustration of the main concept

Recall: β -smooth: GD: $\frac{1}{\eta}$
 LB: $\frac{1}{\eta^2}$

$$\text{GD: } x_{t+1} = x_t - \eta \nabla f(x_t)$$

$$\beta\text{-smooth; } \alpha\text{-sc: GD: } \left(\frac{\kappa-1}{\kappa+1}\right)^T$$

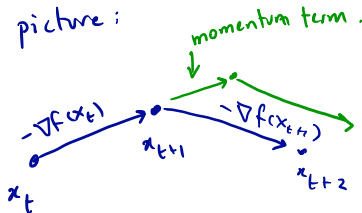
Idea of momentum: $(x_t - x_{t-1})$

$$\text{LB: } \left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^T$$

GD + Momentum:

$$\kappa = \beta/\alpha.$$

Idea by picture:



Acceleration: Smooth and strongly convex functions

min: $f(x)$, f is β -smooth and α -strongly convex

$$\text{GD: } x_{t+1} = x_t - \eta \nabla f(x_t), \quad \eta = \frac{1}{\beta}.$$

AGD: Initialize: $x_1 = y_1 = x_{\text{init}}$

$$y_{t+1} = x_t - \frac{1}{\beta} \nabla f(x_t)$$

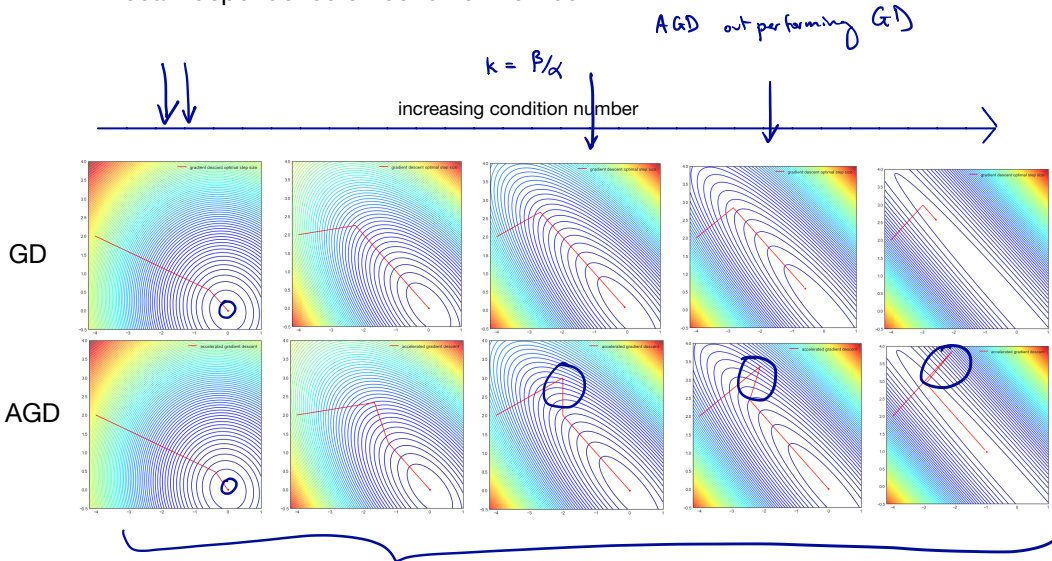
$$x_{t+1} = \left(1 + \frac{\sqrt{k}-1}{\sqrt{k}+1}\right) y_{t+1} - \left(\frac{\sqrt{k}-1}{\sqrt{k}+1}\right) y_t$$

Thm: $f(y_t) - f(x^*) \leq \frac{\alpha + \beta}{2} \|x_1 - x^*\|^2 \cdot \exp\left\{-\frac{(t+1)}{\sqrt{k}}\right\}$

We have improved from $k \mapsto \sqrt{k}$.

Acceleration vs Gradient descent: an example

Recall: dependence on condition number



Acceleration — smooth functions

Gradient : $x^+ = x - \eta \nabla f(x)$ ⑦

Momentum: $d_t = \gamma_t (x_t - x_{t-1})$

GD Alg: $x_{t+1} = (x_t)^+$

Nesterov Acceleration: $x_{t+1} = (x_t + d_t)^+$
 $= x_t + d_t - \eta \nabla f(x_t + d_t)$

f is convex & β -smooth.

Will show we match LB: $1/T^2$ compared to $1/T$

Acceleration for smooth functions — proof of conv rate

Exposition due to S. Bubeck

$$\textcircled{*} \quad x_{t+1} = (x_t + d_t)^+ = (x_t + d_t) - \eta \nabla f(x_t + d_t), \quad d_t = \gamma_t (x_t - x^*)$$

f is β -smooth & convex :

with $\eta = 1/\beta$:

$f(x) - f(y) \leq \langle \nabla f(x), x - y \rangle$

convexity

$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\beta}{2} \|y - x\|_2^2$

with $\eta = 1/\beta$:

$f(x^*) - f(x) \leq -\frac{1}{2\beta} \|\nabla f(x)\|_2^2$

Notation: $\delta_t = f(x_t) - f(x^*) \geq 0$

$$g_t = -\frac{1}{\beta} \nabla f(x_t + d_t) : x_{t+1} = x_t + d_t + g_t \quad \textcircled{*}$$

$$\textcircled{a} \quad \delta_{t+1} - \delta_t \leq -\frac{\beta}{2} (\|g_t\|^2 + 2g_t^T d_t)$$

$$\textcircled{b} \quad \delta_{t+1} \leq -\frac{\beta}{2} (\|g_t\|^2 + 2g_t^T (x_t + d_t - x^*))$$

Proof of acc. grad. desc. for ρ -smooth.

$$\textcircled{a} \quad \delta_{t+1} - \delta_t \leq \underbrace{-\frac{\beta}{2} (\|g_t\|^2 + 2 g_t^\top d_t)}_{\left[g_t = -\frac{1}{\rho} \nabla f(x_t + d_t) \right]}$$

$$\rightarrow -\frac{\beta}{2} \left(\frac{1}{\rho^2} \|\nabla f(x_t + d_t)\|_2^2 - \frac{2}{\beta} \nabla f(x_t + d_t)^\top d_t \right)$$

$$= -\frac{1}{2\beta} \|\nabla f(x_t + d_t)\|_2^2 + \nabla f(x_t + d_t)^\top d_t$$

what did we use here?

$$\geq f(x_t + d_t - \eta \nabla f(x_t + d_t)) - \cancel{f(x_t + d_t)}$$

$$+ \cancel{f(x_t + d_t)} - f(x_t) \leftarrow \text{This is LHS of } \textcircled{a}$$

$$\textcircled{a} \quad \delta_{t+1} - \delta_t \leq -\frac{\beta}{2} (\|g_t\|^2 + 2g_t^T d_t)$$

$$\textcircled{b} \quad \delta_{t+1} \leq -\frac{\beta}{2} (\|g_t\|^2 + 2g_t^T (x_t + d_t - x^*))$$

$$(\lambda_{t+1} - 1) \cdot \textcircled{a} + 1 \cdot \textcircled{b} \leq -\frac{\beta}{2} \left[\underbrace{\frac{1}{\lambda_t} (\|x_t + \lambda_t d_t - x^* + \lambda_t g_t\|_2^2)}_{\frac{2}{\beta} u_{t+1}} - \underbrace{\|x_t + \lambda_t d_t - x^*\|_2^2}_{\frac{2}{\beta} u_t} \right]$$

We would like for the RHS to be telescoping;

want: $x_t + \lambda_t d_t - x^* + \lambda_t g_t = x_{t+1} + \lambda_{t+1} d_{t+1} - x^*$

(recall: $d_{t+1} = \gamma_{t+1}(d_t + g_t)$)

$$= x_{t+1} + \lambda_{t+1} \gamma_{t+1} (d_t + g_t) - x^*$$

$$= x_t + d_t + g_t + \lambda_{t+1} \gamma_{t+1} (d_t + g_t) - x^*$$

$$\lambda_t (d_t + g_t) = (\lambda_{t+1} \gamma_{t+1} + 1) (d_t + g_t) \quad \gamma_t \text{ is a parameter we choose}$$

choose γ_t as a function of λ_t to force this to hold.

We have : (a) $\delta_{t+1} - \delta_t$ (b) δ_{t+1}

$$(\lambda_t - 1) \textcircled{a} + 1 \cdot \textcircled{b} \leq \frac{1}{\lambda_t} \quad \left\{ \quad \right\}$$

$$\Rightarrow \lambda_t^2 \cdot \delta_{t+1} - (\lambda_t^2 - \lambda_t) \delta_t \leq \underbrace{u_t - u_{t+1}}$$

now, pick λ_t to make this telescope.

In prev. page, picked δ_t to make this telescoping.

$$\lambda_t^2 \cdot \delta_{t+1} - (\lambda_t^2 - \lambda_t) \cdot \delta_t$$

$\Downarrow$

Need: $\lambda_t^2 - \lambda_t = \lambda_{t-1}^2$: $\lambda_t \propto t$ will satisfy this

specifically: $\lambda_t = \frac{-1 \pm \sqrt{1 + 4\lambda_{t-1}^2}}{2}$

Sum both sides;

Recall: $\delta_t = f(x_t) - f(x^*)$

$$\lambda_T^2 \delta_{T+1} - \lambda_0^2 \delta_1 \leq u_0 - u_T \leq u_0$$

$$\Rightarrow \boxed{\delta_{T+1} \leq \frac{\text{constant}}{\lambda_T^2}} \Rightarrow f(x_t) - f(x^*) \leq \sim \frac{1}{T^2}$$

Summary:

1. Lower bounds for smooth convex opt
are matched by GD + Momentum.
2. Choose coeff. carefully —